

MUNICH INTL, Germany

WMO: 108660

Lat: 48.3478N

Lon: 11.8133E

Elev: 1487

StdP: 13.92

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 8.8	(c) 14.3	(d) 5.0	(e) 7.5	(f) 10.5	(g) 10.5	(h) 9.9	(i) 16.3	(j) 33.9	(k) 47.4	(l) 29.8	(m) 44.3	(n) 4.7	(o) 80	(p) 0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.4	(c) 86.2	(d) 66.8	(e) 82.6	(f) 65.7	(g) 79.2	(h) 64.6	(i) 68.5	(j) 80.9	(k) 67.2	(l) 79.1	(m) 65.8	(n) 76.8	(o) 7.1	(p) 80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	96.3	72.5	62.9	90.8	70.8	62.3	88.7	70.3	33.7	81.7	32.5	78.7	31.5	76.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.1	(b) 21.1	(c) 18.2	(e) 1.8	(f) 90.9	(g) 6.5	(h) 3.4	(i) -2.9	(j) 93.4	(k) -6.7	(l) 95.4	(m) -10.4	(n) 97.3	(o) -15.2	(p) 99.8		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			1.3	72.0	6.5	2.0	-3.3	73.4	-7.1	74.5	-10.7	75.6	-15.4	77.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	31.2	33.5	40.7	48.4	56.4	62.7	65.8	65.1	56.8	49.1	39.5	33.4
	DBStd	14.09	8.73	8.85	7.36	6.91	6.27	6.37	5.36	5.44	5.79	6.66	7.09	7.87
	HDD50	2411	583	462	296	110	15	1	0	0	9	99	322	514
	HDD65	6185	1047	881	754	499	273	119	56	67	252	493	766	978
	CDD50	1912	1	1	7	61	214	381	490	468	212	71	6	0
	CDD65	214	0	0	0	0	8	49	81	70	6	0	0	0
	CDH74	2406	0	0	0	13	136	531	863	770	90	3	0	0
	CDH80	630	0	0	0	1	14	124	253	231	7	0	0	0
(14) Wind	WSAvg	7.5	8.6	8.5	9.3	7.9	7.4	6.6	6.8	6.1	6.3	6.6	7.5	8.2
(15) Precipitation	PrecAvg	33.0	2.0	1.7	2.2	2.0	3.5	4.1	4.2	3.7	2.8	2.3	2.3	2.2
	PrecMax	40.2	5.6	3.5	5.3	4.3	5.9	6.8	8.6	6.6	5.4	5.2	4.7	3.9
	PrecMin	22.1	0.1	0.4	0.6	0.3	0.5	1.8	0.9	1.5	0.9	0.4	0.0	0.2
	PrecStd	3.7	1.1	0.9	1.2	1.1	1.4	1.5	1.7	1.5	1.1	1.1	1.1	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	59.0	66.2	75.4	82.2	88.1	90.0	91.4	80.7	73.0	62.6	55.1
		MCWB	47.8	47.8	51.6	56.7	63.6	68.0	68.0	66.9	64.4	60.0	53.9	48.1
	2%	DB	51.6	53.5	61.1	71.3	78.4	83.9	86.2	86.0	76.8	68.0	57.4	50.3
		MCWB	45.8	45.7	49.3	55.2	62.1	67.3	66.8	66.7	63.5	58.8	51.2	45.9
	5%	DB	46.8	49.9	57.1	66.6	73.7	80.4	82.4	82.1	73.1	64.2	53.4	47.9
		MCWB	42.8	43.9	48.0	53.3	59.8	65.3	65.7	65.7	62.0	56.6	48.7	44.1
	10%	DB	43.0	46.1	53.3	62.7	69.8	76.8	78.9	78.5	68.4	60.5	49.9	44.5
		MCWB	40.4	41.6	46.3	51.1	57.5	63.8	65.0	64.5	59.9	55.2	46.8	42.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.4	49.0	53.3	59.0	65.5	71.0	70.6	70.4	66.5	61.6	54.6	49.3
		MCDB	53.9	56.2	62.0	72.8	77.5	83.3	84.4	83.2	76.3	69.6	60.8	53.2
	2%	WB	46.1	46.5	50.5	56.3	63.1	68.1	68.3	68.2	64.4	59.2	51.7	46.8
		MCDB	51.0	52.8	58.5	67.7	74.8	79.6	81.0	80.8	74.6	66.0	56.2	50.7
	5%	WB	43.0	44.0	48.7	54.1	61.0	66.3	67.1	66.8	62.7	57.5	49.5	44.1
		MCDB	46.2	48.8	55.6	64.2	71.4	77.3	79.0	78.9	70.9	63.0	52.9	46.9
	10%	WB	40.3	41.5	46.7	52.3	58.9	64.6	65.6	65.3	60.9	55.4	47.0	41.5
		MCDB	42.8	45.1	52.9	61.4	67.8	74.5	76.3	76.1	67.6	59.9	49.6	43.7
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	13.7	16.2	18.9	18.6	18.7	19.4	19.6	18.1	15.8	11.2	10.0
		MCDBR	15.5	20.4	25.6	28.4	26.7	26.3	27.3	27.7	25.9	23.2	18.7	15.3
		MCWBR	11.3	13.7	15.4	15.0	12.9	11.9	11.0	11.1	13.5	14.4	13.2	11.4
	5% WB	MCDBR	14.4	18.7	22.6	25.1	24.0	23.5	23.8	24.6	22.7	21.1	17.4	14.0
		MCWBR	11.0	13.0	14.3	14.0	12.1	11.2	10.3	10.8	12.4	13.7	12.8	10.9
(40) Clear-Sky Solar Irradiance	taub		0.301	0.325	0.359	0.393	0.402	0.406	0.411	0.405	0.380	0.351	0.323	0.297
	taud		2.431	2.371	2.327	2.260	2.269	2.293	2.274	2.311	2.355	2.431	2.466	2.457
	Ebn at Noon		245	263	270	271	273	272	268	265	260	250	234	232
	Edh at Noon		21	27	34	40	41	40	41	37	32	26	20	18
(44) All-Sky Solar Radiation	RadAvg		328	609	961	1365	1585	1767	1701	1491	1072	674	355	260
	RadStd		47	82	119	179	184	154	155	150	133	78	45	33

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.65	+0.67	N/A	N/A	N/A	+36
(47) Regional (83 neighbors)	+0.74	N/A	N/A	N/A	+0.58	+0.59	N/A	N/A	+112	+43

Nomenclature: See separate page